

Problem Description

- Manual parking allocation causes delays, incorrect occupancy information and inefficient use of available slots.
- Parking operators need a centralized database to manage lots, slots, vehicles, parking sessions and billing.
- The proposed system automates slot allocation, session tracking and bill generation.

Existing Problems

- Drivers may be assigned slots manually without considering slot type or current availability.
- Paper/manual records make it difficult to know which vehicles are currently parked.
- Billing can be inconsistent when rates, grace periods and parking duration are not centrally maintained.

Proposed Solution

- Store parking infrastructure and vehicle information in a relational database.
- Use stored procedures to allocate the first suitable free slot and process vehicle exits.
- Use triggers and reporting views to maintain slot status and provide occupancy/revenue information.

Objectives and Scope

- Support multiple parking lots and different slot/vehicle types.
- Maintain parking sessions from entry to exit and calculate charges using a rate card.
- Provide a database foundation that can be connected to a user interface for CRUD operations.

Expected Outcome

- A consistent database for parking allocation and billing.
- Automatic occupancy updates through triggers.
- Queryable reports for occupancy, active sessions and daily revenue.